

Introduction

Datasets from different batches, donors, or technologies cluster by batch rather than biology without correction. Integration methods align shared cell types across datasets while preserving dataset-specific populations. Harmony iteratively centres cell embeddings by batch; Seurat's CCA/RPCA anchors cells between batches.

Prerequisites

Multiple scRNA-seq datasets with shared cell populations; normalised expression; PCA embeddings.

Theory

Harmony alternates (i) soft cluster assignment in PCA space, (ii) per-cluster batch centroid computation, (iii) shifting cells within each cluster by its batch-specific correction vector. Converges in a few iterations and scales to millions of cells.

Seurat CCA/RPCA integration finds anchor pairs across datasets via mutual nearest neighbours in a shared low-dimensional space and uses anchors to transform embeddings.

Assumptions

Shared cell types span batches; batch effects are corrective in PCA space; biology and batch are not perfectly confounded.

R Implementation

```
library(harmony); library(Seurat)

set.seed(2026)
# Simulate two batches with shared biology + batch shift
n <- 150
batch <- rep(c("A", "B"), each = n)

counts <- matrix(rpois(300 * 200, lambda = 2), 200, 300)
rownames(counts) <- paste0("g", 1:200)
colnames(counts) <- paste0("c", 1:300)
counts[, batch == "B"] <- counts[, batch == "B"] + 3 # batch effect

so <- CreateSeuratObject(counts)
so$batch <- batch
so <- NormalizeData(so, verbose = FALSE)
so <- FindVariableFeatures(so, nfeatures = 100, verbose = FALSE)
so <- ScaleData(so, verbose = FALSE)
so <- RunPCA(so, npcs = 10, verbose = FALSE)

so <- RunHarmony(so, group.by.vars = "batch", dims.use = 1:10,
                 verbose = FALSE)
so <- RunUMAP(so, reduction = "harmony", dims = 1:10, verbose = FALSE)
so <- FindNeighbors(so, reduction = "harmony", dims = 1:10, verbose = FALSE)
so <- FindClusters(so, resolution = 0.3, verbose = FALSE)

table(Batch = so$batch, Cluster = Idents(so))
```

Output & Results

A Harmony-corrected embedding; after clustering, batches should be represented in each cluster (shared cell types) rather than segregating by batch.

Interpretation

“After Harmony integration, all seven major cell types contained cells from both donors (mixing LISI = 1.84 / 2.0), indicating successful batch correction without over-mixing.”

Practical Tips

- Use Harmony for speed; CCA/RPCA for heavier anchor-based correction when batches have unique cell types.
- Quantify integration quality with kBET, LISI, or silhouette by batch vs cell type.
- Never integrate across tissues; integrate within tissue with donors/batches as covariates.
- Over-integration destroys biology; always inspect cell-type marker expression post-integration.
- Batch as a fixed effect in the DE model is an alternative to explicit embedding integration for focused comparisons.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- RNA-seq with DESeq2 / edgeR
- Enrichment analysis (GSEA / ORA)
- scRNA-seq with Seurat